Prem Lodhia

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PROFILE

First-year Computer Science with Artificial Intelligence student building hands-on capability in cloud security, application security, and detection engineering. Developed local security labs covering web exploitation and remediation, log-based detection, alert triage, and rule tuning. Comfortable with Python, Bash, PowerShell, Docker, IAM/networking fundamentals, and writing clear technical documentation.

SELECTED PROJECTS

BlueTrace Lab | Detection Engineering Lab

- Built a local detection engineering lab that simulates authentication, process, file-access, and network events, normalizes logs, and applies rule-based detections to generate triage-ready alerts.
- Added replayable suspicious scenarios, MITRE ATT&CK mapping, alert severity/status tracking, and tuning logic to show how noisy detections can be improved without losing useful signal.
- Implemented the lab with Python, SQLite, Docker, and a lightweight dashboard, then documented the workflow from event generation to alert review.

VulnShop Lab | Local Web Application Security Lab

- Built a local-only e-commerce lab with side-by-side vulnerable and remediated builds so insecure and secure behaviour can be compared directly.
- Tested and documented issues including SQL injection, XSS, IDOR, SSRF, insecure file upload handling, weak session controls, brute-force exposure, validation gaps, and error disclosure.
- Produced structured findings, remediation, and retest documentation; implemented the lab with Node.js, Express, SQLite, and Docker.

Cyber Security Toolkit (Python) | CLI Security Utilities

- Built a modular Python toolkit covering password strength checks, hashing and cracking demonstrations, TCP port scanning, brute-force log analysis, and encryption/decryption utilities.

EXPERIENCE

Outlier | AI / Data Work

- Completed detail-oriented AI and data tasks that strengthened pattern recognition, anomaly spotting, written reasoning, and disciplined quality control.

Birmingham City University | IT Support Practice

- Supported users with troubleshooting and issue resolution, building practical experience in diagnosing access issues, explaining fixes clearly, and maintaining reliable day-to-day operations.

TECHNICAL SKILLS

Languages & Automation: Python, Bash, PowerShell, SQL, Git/GitHub
Cloud & Infrastructure: AWS fundamentals (IAM, EC2, S3, Lambda, VPC), Azure fundamentals, Docker, Terraform (basic)
Security: OWASP Top 10, least privilege/IAM, detection engineering fundamentals, CloudTrail, Splunk, Microsoft Sentinel concepts, Nmap, Wireshark, Nessus, Qualys
Networking: TCP/IP, DNS, HTTP/HTTPS, subnetting, firewalls, VPN, TLS/SSL

EDUCATION

Birmingham City University | BSc Computer Science with Artificial Intelligence
First-year student building hands-on capability through coursework, labs, and security project development.

George Dixon Academy | A Levels
Biology, Chemistry, Physics

ADDITIONAL

Captained a cricket team, building leadership, teamwork, communication, and decision-making under pressure. Active on GitHub and LinkedIn while continuing to expand practical knowledge in cloud, cybersecurity, and secure software engineering.